

Practical 10 - Estimating causal effects using observational data 1

Survival Analysis

London School of Hygiene & Tropical Medicine

Stata version (please see accompanying Stata do file)

SOLUTIONS

Datasets required: `rotterdam_p10`

Introduction

In this practical you will perform analyses to obtain the results shown in the lecture, using the Rotterdam breast cancer data. The Rotterdam data comprises data on 2982 individuals who underwent surgery for primary breast cancer and whose data were recorded in the Rotterdam Tumour Bank. Participants were followed-up for up to 20 years for the outcomes of disease recurrence and death from any cause. In this practical the focus is going to be on the composite outcome of disease recurrence or death without recurrence, and on the first 5 years of follow-up. The data have been modified from the original to apply administrative censoring at 5 years.

The variables we will use are summarised in the table below.

Variable name	Description
<code>pid</code>	Patient identifier
<code>year_cat</code>	Year of surgery. 0: before 1990, 1: 1990 or later
<code>age</code>	Age at surgery
<code>meno</code>	Menopausal status. 0=premenopausal, 1=postmenopausal
<code>size</code>	Tumour size: categorical with levels ≤ 20 mm, 20 – 50mm, > 50 mm

Variable name	Description
<code>grade</code>	Differentiation grade: categorical with levels 2 and 3
<code>enodes</code>	Number of positive lymph nodes, transformed as $\text{enodes} = \exp(-0.12 * \text{nodes})$
<code>pgr</code>	Progesterone receptors (fmol/l)
<code>er</code>	Estrogen receptors (fmol/l)
<code>chemo</code>	Chemotherapy: 0=no, 1=yes
<code>hormon</code>	Hormonal treatment: 0=no, 1=yes
<code>time</code>	Years to the first of recurrence or death without recurrence
<code>outcome</code>	Indicator of composite outcome: 0=censored, 1=event
<code>outcome_cr</code>	Competing events outcome: 0=censored, 1=recurrence, 2=death without recurrence

Aims

The aims of this session are:

- Estimate conditional and marginal survival probabilities at a specific time under different treatment options.
- Estimate causal effects of a treatment in terms of marginal risk differences, risk ratios and restricted mean survival times.
- Estimate marginal survival curves under different treatment options.

Question 1

- Load the data.
- How many events are there and how many used hormone therapy?
- Compare the characteristics of users and non-users of hormone therapy.

SOLUTIONS

- Among the 2982 people, there are 1285 events in the first 5 years of follow-up, with the remaining 1697 individuals being censored.
- 339 people used hormone therapy.
- Hormone therapy users tend to be older, more likely to be post-menopausal, tended to have larger tumour size and higher grade, tended to have lower values for pgr but higher values for er, and were less likely to have chemotherapy. Hormone therapy use was also more common in the post-1990 period.

Question 2

- Use `stset` using `time` and `outcome`.
- Fit an unadjusted Cox model including only the hormone therapy variable. Interpret the hazard ratio estimate.
- Fit a Cox model with hormone therapy as your main exposure, and conditioning on all other explanatory variables included in the data. The variables to include are: `age`, `meno`, `size`, `grade`, `enodes`, `pgr`, `er`, `year_cat`, `chemo`. We will assume that these are all potential confounders. Interpret the hazard ratio estimate for the hormone therapy variable and compare it with the unadjusted estimate.

SOLUTIONS

- In the unadjusted model, the HR for hormone therapy is 1.27 (95% CI 1.08 to 1.49). In other words, hormone therapy use is associated with a 27% higher hazard for the composite event of recurrence or death (95% CI 8% to 49%).
- After conditioning on the set of variables listed above, the HR for hormone therapy is 0.73 (95% CI 0.61 to 0.88). That is, hormone therapy use is associated with a lower hazard for the composite event of recurrence or death after conditioning on the potential confounders.

Question 3

In this question we will use the adjusted model fitted in Question 2 to obtain an estimate of the risk of the composite event up to 5 years for each individual in the data under two treatment strategies:

- (i) Start hormone therapy following surgery (and continue as recommended by the doctor) (`hormon=1`),
- (ii) Do not start hormone therapy following surgery (`hormon=0`).

Use the code in the do file to do this for (i), and modify it for (ii).

What is the estimated risk of recurrence or death up to 5 years for person 1 in the data if they were to have followed treatment strategies (i) and (ii)?

SOLUTIONS

For an individual with the same characteristics as person 1 in the data, except for their hormone therapy status, the risk of the composite event up to 5 years is estimated to be 20% if they were to take hormone therapy and 26% if they were not to take hormone therapy.

Question 4

In this question we will use the results from the previous question to obtain estimates of the marginal (adjusted) risks under the two treatment strategies.

- Obtain the marginal risk estimates $\Pr(T \leq 5|do(X = 1))$ and $\Pr(T \leq 5|do(X = 0))$, where X denotes hormone therapy strategy and T denotes the time of the composite event of interest. This can be done by taking the average of the conditional risk estimates obtained in question 3.
- Obtain estimates of the risk difference $\Pr(T \leq 5|do(X = 1)) - \Pr(T \leq 5|do(X = 0))$ and the risk ratio $\Pr(T \leq 5|do(X = 1)) / \Pr(T \leq 5|do(X = 0))$.

SOLUTIONS

- If everyone had taken hormone therapy the expected marginal (population average) probability of recurrence or death up to 5 years is 36.4%. If everyone had not taken hormone therapy the expected marginal (population average) probability of recurrence or death up to 5 years is 44.8%.
- The marginal risk difference (hormone therapy versus no hormone therapy) is -8.4%. If everyone had taken hormone therapy the risk of recurrence or death within 5 years would have been 8.4 percentage points lower than if everyone had not taken hormone therapy.
- The marginal risk ratio is 0.81. If everyone had taken hormone therapy the risk of recurrence or death within 5 years would have been 19% lower than if everyone had not taken hormone therapy.

Question 5

- In this question we will use a ready-made function `standsurv` to estimate the marginal (adjusted) risks differences and risk ratios, alongside confidence intervals. Take a look at the `do` file to do this. This only uses fully parametric models, so we do it using a Weibull model in place of the Cox model used above.
- Find the marginal risks and the marginal risk difference and risk ratio, and their 95% confidence intervals

SOLUTIONS

The estimates are shown in the table below, based on the Cox model and based on the Weibull model. The results are almost identical.

Estimand	Estimate	95% CI
Based on Cox model		
Marginal risk up to 5 years: <code>hormon=1</code>	0.36	0.32 to 0.41
Marginal risk up to 5 years: <code>hormon=0</code>	0.45	0.43 to 0.47
Marginal risk difference	-0.084	-0.127 to -0.040
Marginal risk ratio	0.81	0.72 to 0.91
Based on Weibull model		
Marginal risk up to 5 years: <code>hormon=1</code>	0.37	0.33 to 0.42
Marginal risk up to 5 years: <code>hormon=0</code>	0.46	0.44 to 0.47
Marginal risk difference	-0.085	-0.131 to -0.038
Marginal risk ratio	0.81	0.72 to 0.92

Question 6

So far we have focused on estimating the effect of hormone therapy on the risk up to time 5. We now extend this to obtain estimates of the marginal adjusted survival curves under the two treatment strategies.

- See the do file for how to do this using `standsurv`.
- What is the interpretation of the resulting curves?

SOLUTIONS

The curves are the estimated survival curves (for recurrence-free survival) for this population if everyone had used hormone therapy and if everyone had not used hormone therapy.

Extra questions/examples

Question 7

In the above analyses we obtained marginal risks (and risk differences and risk ratios) based on standardisation of conditional estimated from a Cox model that made the proportional hazards assumption for all variables included in the model.

In this question we relax the proportional hazards assumption for `hormon` by stratifying on this variable, i.e. by allowing a separate baseline hazard in the `hormon=0` and `hormon=1` groups. After fitting this modified model, the subsequent standardisation steps are the same as above.

See the code below for how to do this, and compare the results with those from Question 5.

SOLUTIONS

The estimates are shown in the table below. The results based on the stratified Cox model are very similar to those based on the model assuming proportionality of hazards for hormone therapy.

Estimand	Estimate	95% CI
Cox model assuming proportional hazards for hormon		
Marginal risk up to 5 years: hormon=1	0.36	0.32 to 0.41
Marginal risk up to 5 years: hormon=0	0.45	0.43 to 0.47
Marginal risk difference	-0.084	-0.127 to -0.040
Marginal risk ratio	0.81	0.72 to 0.91
Cox model stratified by hormon		
Marginal risk up to 5 years: hormon=1	0.37	0.32 to 0.41
Marginal risk up to 5 years: hormon=0	0.45	0.43 to 0.47
Marginal risk difference	-0.081	-0.130 to -0.031
Marginal risk ratio	0.82	0.71 to 0.93

Question 8

Another estimand of interest could be the restricted mean survival times (RMST) under the two treatment strategies, or differences in the RMST. Based on the estimates of the marginal risks (or survival probabilities) obtained above, here we show how to estimate the RMST up to time 5 under the two treatment strategies, and their difference.

The code provided illustrates how to estimate the marginal RMST and RMST differences using `adjustedsurv`.

Interpret the RMST estimates and their difference.

SOLUTIONS

The RMST is interpreted as the expected time alive and free of disease recurrence up to 5 years. The estimated RMST if everyone were to use hormone therapy is 4.04 years, and estimated RMST if everyone were not to use hormone therapy is 3.79 years. The RMST difference between the two treatment strategies is 0.26 (95% CI 0.12 to 0.40). This means that if everyone had not used hormone therapy, the expected time alive and free of disease recurrence up to 5 years would be 0.26 years less than if everyone had used hormone therapy.